SIU Southern Illinois University CARBONDALE

PHYSICS DEPARTMENT

Mark Byrd
Physics Department
Southern Illinois University
mbyrd@siu.edu

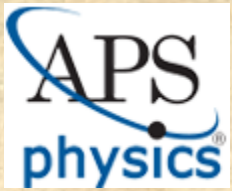
I graduated from the University of Texas at Austin with a PhD in Physics
(I also have undergraduate degrees in physics and mathematics).
Advisors: E.C.G. Sudarshan and Duane Dicus.

Taught one semester at Southern Methodist University in Dallas, Texas

Postdoctoral fellow in the **Chemistry Department** at the University of Toronto with Daniel Lidar.

Postdoctoral fellow at Harvard University in the **Division of Engineering and Applied Science** where I worked with an **applied mathematician** (Navin Khaneja) who is a **Professor of Electrical Engineering**.

Now faculty in the Physics Department at Southern Illinois University



Quantum Information (GQI)
American Physical Society
TOPICAL GROUP ON QUANTUM INFORMATION:

Topical Group on Quantum Information

The mission of the Topical Group on Quantum Information is to promote the advancement and diffusion of knowledge concerning the physics of quantum information, computing, fundamental concepts, and foundations. The Topical Group will serve as a focus for theoretical and experimental research in these and related areas. Research topics of direct interest include quantum entanglement, quantum communication, quantum cryptography, quantum algorithms and simulations, physical implementations of qubits, quantum error correction, fault-tolerant quantum computation, quantum measurements, open quantum systems, quantum coherence, control of quantum dynamics, the quantum-classical correspondence, and the conceptual and mathematical foundations of quantum theory.

Join the APS: Find out about Careers, Opportunities in Academia, etc.
http://www.aps.org/membership/join.cfm

Join the GQI:
http://goo.gl/forms/yzJyJJf5gB

- What do Physicists do? Teaching?

  Materials science, semiconductor industry,
  energy control/production,
  laser/light technology, optics,
  medicine (radiation dosages, NMR/MRI), →
  traffic control, →
  computer science, →
  electronics,
  space physics,
  defense,
  finance,
  patent law, →
  geophysics,
  oil exploration, →
  communications research/development, … →

  also teaching

Why Study Physics? Future of a Physicist with a BS.

Employers in Illinois that recently hired new physics bachelor recipients

Accenture
Aerotek Scientific
Aisin Electronics Illinois, LLC
Analysts, Inc.
Argonne National Lab
Army Corps of Engineers
Beckman Institute (University of Illinois)
Caterpillar, Inc.
CONTAX, Inc.
Creative Thermal Solutions
CSG Systems
Delcross Technologies, LLC
Deloitte Consulting, LLP
Exelon
Fermi National Lab
Fuji Machine America Corporation
GMS / Vedior
Greenlight Planet, Inc.
Highland Engineerin, P.C.
Imaje
Leo Burnett
Lockheed Martin
Northrop Grumman
Orchid Tree WEb Solutions
Quantum Design
S&C Electric Company
Sargent & Lundy
United Conveyor Corporation
Val-Matic Valve & Manufacturing Corporation
Wellpoint, Inc.
Zurich North AmericaS&C



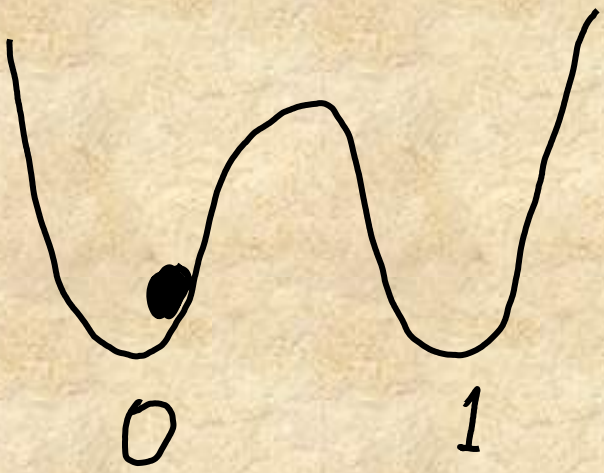
This is only a portion of the employers w ho hired recent physics bachelors into technical positions.
Source: AIP Statistical Research Center, Initial Employment Surveys, classes 2007 thru 2009.

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Quantum Mechanics

First we should note that information is stored in physical systems and although we very often observe systems which behave classically. These systems obey Newton's Laws. However, quantum mechanics is a more fundamental theory and governs the behavior of small things.

Consider the following problem:



The state of the particle can be described by $|0\rangle$ if it's in the first well and $|1\rangle$ if it's in the second well. If it is in a superposition, (in some sense both wells at once) we can describe it as

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle,$$

where α and β are complex numbers and $|\alpha|^2 + |\beta|^2 = 1$

We call quantum two-state systems quantum bits, or qubits.

Quantum Computing/Quantum Information

.***What is quantum information?***

Storing and manipulating quantum systems for the purposes of information storage, manipulation, and transfer.

<u>Why should we care</u>? ***THIS IS THE INFORMATION AGE!! But also …***

"How can we resist probing this quantum world, which is as exciting and surprising as a newly discovered planet orbiting a distant star? It's serious business too, if 60 percent of <US> GDP depends on the mastery of quantum science."

Quantum Physics for Poets**,** By Leon M. Lederman, Christopher T. Hill

A list of modern technologies that arose from quantum mechanics would include: "transistors, microprocessors, lasers, sophisticated chemistry and materials science, nuclear energy, memory chips, hard drive storage, LEDs, LCD displays, etc."

-Stephen Hsu

<u>What I do:</u>

Much of my work concerns preventing errors in quantum information processing devices since errors are the greatest obstacle to building a scalable quantum computing device.

I also study basic open-systems in quantum mechanics and quantum information.

Quantum Computing: Why?

1) To make computers faster and more compact, we have been making them smaller. (*Moore's Law*) There is a limit to small … quantum mechanics starts to become the dominate mechanism by which constituents interact.

So, to make things smaller, we need to use quantum mechanics!

2) One may also suppose that when the physical mechanisms which we use for components obey different laws, the machine will behave differently – that is indeed the case!

3) We now know of several different quantum algorithms which are faster than any known classical algorithm.

4) Quantum information can be used in a variety of ways beyond computing. Such as quantum cryptography, quantum games, and quantum communication of all sorts.

Quantum Computing: Advantages

There are three advantages of quantum computing devices which are often quoted. This is due to the fact that a quantum computer, could

1) Factor large integers more efficiently than any known classical algorithm (called Shor's algorithm)
2) Find an object in an unsorted database more efficiently than a classical machine (known as Gover's algorithm, e.g. phone number lookup)
3) Simulate quantum mechanical systems more efficiently than any known classical method (due to Feynman and others)

COMMENTS:

- Shor's algorithm would render RSA encryption useless. It is more efficient than any known classical algorithm. (There is a quantum answer to this problem however- quantum cryptography through QKD.)

- Gover's algorithm is better than any classical algorithm – phone book example: classical algorithm grows as N/2 and Grover's grows as $\sqrt{N}$

- Simulating quantum mechanical systems is quite difficult classically. For physical scientists this could be the most important application of quantum computers. This could enable the simulation of solid-state devices, biological molecules and molecular interactions, etc.

Quantum Computing: Advantages

NUMBERS >>>>

Factor large integers efficiently:

Classically factoring 1024-bit number: $>10^{10}$ years (2 GHz Machine)

Quantum factoring 1024-bit number: several seconds (1 GHz Machine)

Find an object in an unsorted database:

Classically searching 10^{10} objects: $5X10^{9}$ queries

Quantum searching 10^{10} objects: 10^{5} queries

Simulate quantum mechanical systems:

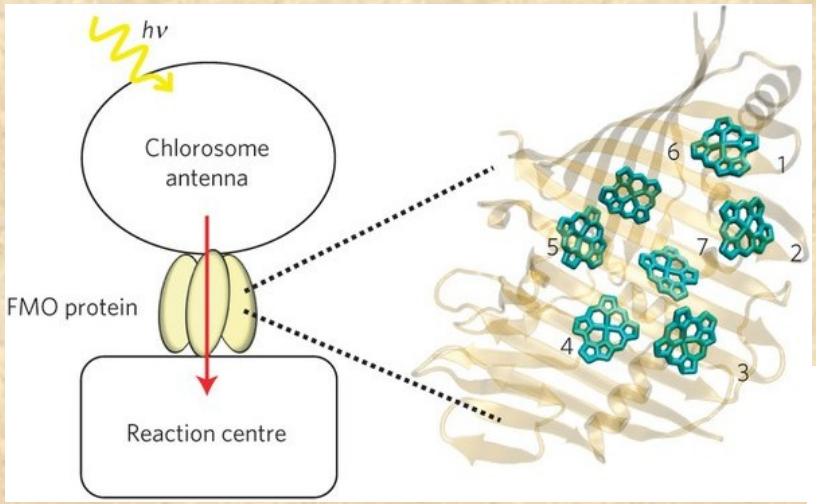
Classically represent Hilbert space » $2^{N}X2^{N}$ matrix

Quantum mechanical, use N particles

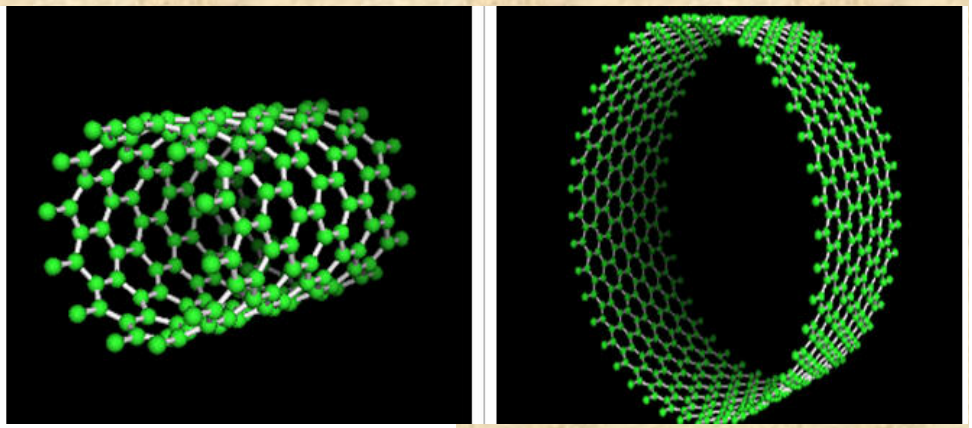
Simulating Quantum Systems

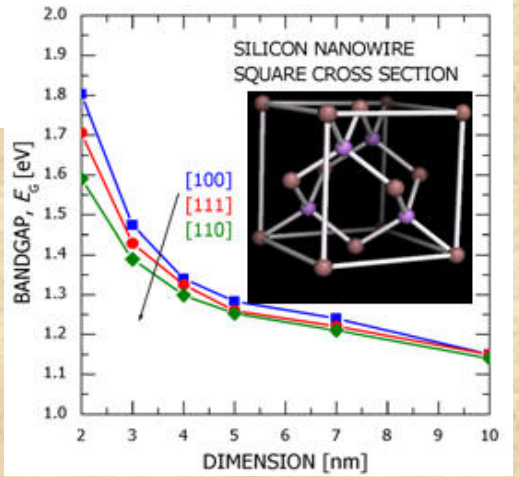
For physical scientists, the latter seems by far the most important. It could enable us to simulated many-body quantum systems and does not require a quantum computer: Chemistry, Biomolecules, Solid-State Systems and Materials

We do not need a fully functioning quantum computing device.



Fenna-Matthews-Olson (FMO) Olson:04 complex is one of the two types of antennae found in green sulphur bacteria.





Shaikh Ahmed, et al.

Also DNA

http://www.engr.siu.edu/staff1/ahmed/mywebpage/Research_Projects_ShaikhAhmed.htm#(A)_THEORY/MODELS/ALGORITHMS_

Quantum Technologies

Quantum Computing

Quantum Simulation

Quantum Cryptography
Quantum Key Distribution
(Enables sharing a secret key for secure communication.)

Quantum Games (cooperative games, economics, etc.)

Quantum Sensing (Electronic Noses for example)

Quantum Imaging (NMR, or MRI, STM, TEM)

Quantum Metrology
(See for example the Quantum Metrology Institute,
http://www.npl.co.uk/about/quantum-metrology-institute)

Quantum Technologies



The Quantum Metrology Institute (QMI) will cover all of NPL's leading-edge quantum science and metrology research and provide the expertise and facilities needed for academia and industry to test, validate, and ultimately commercialise new quantum research and technologies.

By bringing together industry engineers, academic researchers, and NPL scientists in a highly collaborative environment, the QMI will play a key role in the creation of a UK industry based on quantum technologies.

Aim to formally open the QMI in late 2015. The new facility will include a range of new capital investments funded from a £4M capital budget for quantum technologies at NPL, announced as part of the overall government investment in the field.

Quantum Technologies are based on the properties of atoms, electrons and photons (basic units of light) of everyday objects. These fundamental particles can provide unchanging standards to use for measurement and their 'quantum' behaviour can potentially create a completely new class of advanced sensors and systems, with radically new capabilities for timing and navigation, communications, computing and other critical applications.

Which Companies Use Quantum Technologies and For What?

See Quantum Technologies on Wikipedia

Quantum Technology (Frontiers of Science)
by Gerard Milburn and Paul Davies

"Quantum Technology: The Second Quantum Revolution,"
J.P.Dowling and G.J.Milburn, arXiv:quant-ph/0206091v1
(1st was understanding QM, 2nd was using the weird properties.

BBN,
Applied Communication Sciences
HP,
Microsoft,
D-Waves,
MagiQ,
Qubitekk
Sandia National Labs,
HRL,
Laboratory of Physical Sciences,
Google

NPL
Lockheed Martin
NASA
NRL
Mitre
ID Quantique

Quantum Computing: D-Wave Systems

How D-Wave Systems Work

Quantum computing uses an entirely different approach than classical computing.
A useful analogy is to think of a landscape with mountains and valleys.
Solving optimization problems can be thought of as trying to find the lowest point on this landscape. Every possible solution is mapped to coordinates on the landscape, and the altitude of the landscape is the “energy’” or “cost” of the solution at that point.
The aim is to find the lowest point on the map and read the coordinates, as this gives the lowest energy, or optimal solution to the problem.
Classical computers running classical algorithms can only "walk over this landscape".
Quantum computers can tunnel through the landscape making it faster to find the lowest point. The D-Wave processor considers all the possibilities simultaneously to determine the lowest energy ...

Capabilities

D-Wave’s flagship product, the 1000-qubit D-Wave 2X quantum computer, is the most advanced quantum computer in the world.

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Quantum Computing: D-Wave Systems

It is best suited to tackling complex optimization problems such as:

- Optimization
- Machine Learning
- Pattern Recognition and Anomaly Detection
- Financial Analysis
- Software/Hardware Verification and Validation

Places that have purchased a D-Wave processor:

- NASA
- LOCKHEED MARTIN
- GOOGLE

Quantum Computing: D-Wave Systems

Two People I know who work at Google:

Dave Bacon: (at Google over 4 years) Research in Quantum Error Correction
PhD, Physics UC Berkeley
Postdoc Caltech
Postdoc Santa Fe Institute
Discoverer of the code that now bears his name
the "Bacon-Shor" Quantum Error Correcting code

Alireza Shabani: (at Google over 1 year) Research in Quantum Error Correction and Quantum Tomography
PhD, Physics University of Southern California
Postdoc Princeton University
Postdoc UC Berkeley

Google

Good Salary, Benefits, Flexible Schedule

Part of the Job is to do something of your own choosing for 10 hours per week.

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PHYSICS DEPARTMENT

Graduate School

Get paid to go to graduate school in Physics @ SIU

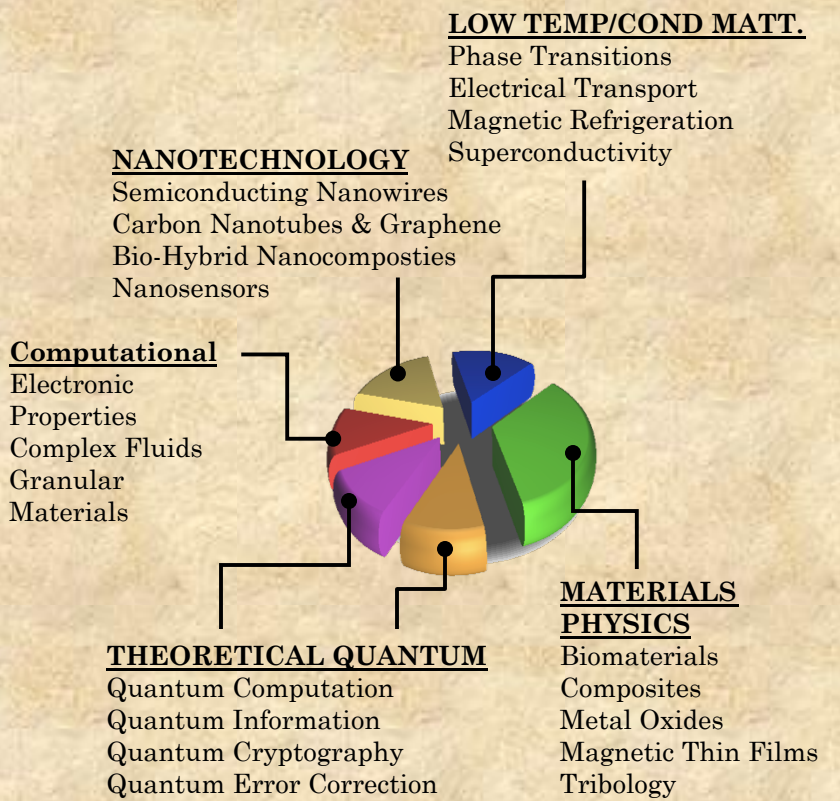
Usually teach 1 or 2 years and then have a research assistantship

Tuition is waved (NO Tuition!)

Experimental research or

Theoretical research

Department Research Topics



Applying to Graduate School at SIU:

http://www.physics.siu.edu/graduate/apply_now.php